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Time-lapse imaging of cytoplasmic strings at the blastocyst stage indicates their association with spontaneous collapse

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Abstract

Research question: To study the origin and temporal behaviour of cytoplasmic strings spanning the blastocoel (main objective) and their influence on treatment outcome (secondary objective).

Design: This retrospective analysis of prospectively collected data was set up in a university medical centre. Patients who either underwent fresh (n = 95) or vitrified/warmed (n = 55) single blastocyst transfer were included. Time-lapse sequences of *in vitro* developed blastocysts were screened for the presence of cytoplasmic strings. Pregnancies in string-positive (group 1) and string-negative (group 2) transfers were followed up to live birth.

Results: A total of 387 blastocysts were achieved in the fresh cycles of 100 patients, corresponding to a blastocyst formation rate of 62.4%. Cytoplasmic strings (tCS) were first detected around full stage (108.5 ± 6.4 h) in 170 blastocysts (43.9%). The number of strings varied (range: 1-7) and the duration of visibility (VCS) was 5.2 ± 3.5 h. The occurrence of cytoplasmic strings was significantly associated with the presence of blastocoelic collapses ($P < 0.001$) but not with any of the annotated morphokinetic parameters. Live birth and neonatal outcome were the same for both string-positive and -negative pregnancies. Moreover, collapses did not affect treatment outcome.

Conclusion: Time-lapse analysis of cytoplasmic strings at the blastocyst stage revealed that this morphological feature was not a negative predictor as previously reported, but could serve as a positive selection criterion of outcome. Although physiologically normal, at least some of the cytoplasmic strings are an artefact possibly associated with blastocoelic collapses.

Key words: blastocoel collapse; blastocyst morphology; cytoplasmic strings; morphokinetics; time-lapse

Introduction

Although invasive techniques, such as trophectoderm biopsy (Capalbo et al., 2016a), are the most precise approaches for estimating the implantation potential of selected embryos, these are no feasible options for most *in vitro* fertilisation (IVF) centres worldwide for several reasons. These centres rather use non-invasive technologies for achieving a healthy birth of a singleton baby, which is the ultimate goal of assisted reproduction.

One way of indirectly predicting treatment outcome would be to use associated cells of somatic origin (Corn et al., 2005; Cheng et al., 2013) or spent culture medium (Capalbo et al., 2016b; Huang et al., 2019; Liang et al., 2019) to estimate the further fate of transferred embryos. However, again the vast majority of clinical embryologists have to rely on less sophisticated techniques.

For non-invasive selection purposes, examining simple static morphology at different stages of preimplantation development is still the most commonly used practice (Ebner et al., 2003), but it seems that embryo scoring morphokinetics (Meseguer et al., 2012) is slowly overtaking pure morphological evaluation. This is not only valid for culturing fresh embryos but also for cryopreserved ones (Ebner et al., 2017a; Kovačič et al., 2018). An expert time-lapse user group recently published a consensus paper on how to annotate dynamic human embryo development based on exact cleavage timings, duration of cell cycles, and synchronicity of these events (Ciray et al., 2014). Time-lapse imaging also helped facilitate the identification of irregular cleavages such as rapid cleavage, cell fusion, and trichotomous mitosis (Rubio et al., 2012; Liu et al., 2014; Almagor et al., 2015; Zjan et al., 2016).

Apart from this positive and negative predicting capacity, time-lapse techniques are particularly suitable to describe new structures (Derrick et al., 2017; Kellam et al., 2017; Coticchio et al., 2018), to evaluate the exact course of certain cellular events (Liu et al., 2015; Ebner et al., 2017b; Coticchio et al., 2018), as well as to re-evaluate findings that were previously published based on morphological analysis alone.

The fact that time-lapse technology enables continuous monitoring of cellular events bears the risk that well-established static parameters perform differently once morphokinetic data is considered. With respect to this, it is quite interesting that the presence of multinucleation (Balakier et al., 2016) was massively underestimated before time-lapse imaging was available (Ergin et al., 2014). Other time-lapse studies, such as those related to the planar arrangement of blastomeres (Ebner et al., 2017b), more or less confirmed what had been hypothesised using morphological data alone (Ebner et al., 2012).

The above-mentioned potential divergence between morphological and morphokinetic results would require an accurate re-evaluation of less prominent morphological features through time-lapse technology to obtain more robust data which could then add to a decision process of clinical relevance.

The presence of cytoplasmic strings at the blastocyst stage, previously described as a negative feature (Scott, 2000), is one such parameter about which limited information is available. Therefore, the present study sought to explore the origin, nature, and temporal behaviour of cytoplasmic strings spanning the blastocoel at the blastocyst stage. To the best of our knowledge, this is the first time-lapse study on string-positive blastocysts and their potential association with live birth.

Material and methods

Before submission of this retrospective dataset, ethical approval was sought and granted (1175/2018) by the Ethical Committee of Upper Austria.

Study design

In the 18-month study period, a total of 510 patients (corresponding to 739 cycles) were assessed for eligibility (Figure 1). One hundred patients fulfilled the inclusion criteria, which included single blastocyst transfer on day 5 after culture in a time-lapse incubator.

The study was split into two phases. The primary objective was to screen all blastocysts available in the study cohort (n = 100) for the presence of cytoplasmic strings, which was our morphological criterion of interest. Since this feature becomes visible before cryopreservation, we also planned to include cycles with vitrified/warmed blastocysts in the case of a freeze-all strategy (n = 5) or no pregnancy. Thus, only those cases were considered whose embryos were vitrified on day 5. Consequently, we ensured that all relevant data on day 5 could be collected for all involved embryos, i.e., both fresh and vitrified. The secondary endpoint then included treatment outcome after either fresh or vitrified/warmed single blastocyst transfer (150 transfers from 100 patients). Patients who had string-positive transfers did not differ from their string-negative counterparts in terms of demographic and stimulation data (Table 1).

Patient characteristics

Mean female age was 33.1 ± 6.3. The couples represented an average IVF cohort with most of them suffering from male factor infertility (82%), either isolated or in combination with female indications like polycystic ovarian syndrome (PCOS, 9%), endometriosis (7%), or tubal infertility (6%) (multiple entries possible). The ratio between primary and secondary infertility was 60:40.

Controlled ovarian hyperstimulation

Since the antagonist protocol is the preferred mode of stimulation at the subject university hospital, the vast majority of cases (82%) was stimulated with recombinant (Puregon, MSD, Vienna, Austria) and/or urinary gonadotropins (Merional or Fostimon, IBSA, Lugano, Switzerland) as well as Ganirelix (MSD). The remaining couples, mainly patients with endometriosis and those who had successful birth after a long protocol, had been administered an GnRH-agonist like Triptorelin (Ferring, Vienna, Austria) or Buserelin (Sanofi-Aventis, Frankfurt, Germany). The type of gonadotropins used was the same in the long protocol.

Routinely, oocytes were collected 36 h after inducing ovulation using 10,000 IU of hCG (Pregnyl, MSD). GnRH agonists were never used to induce ovulation. On an average, stimulation took 10.6 ± 2.3 days and the oestradiol level on the day of ovulation induction was found to be 2,423 ± 790 pg/mL. It should be considered that only those cases are involved in this study which were cultured up to blastocyst stage. These patients normally have more oocytes available.

In vitro culture

After a short resting period of approximately 2 h, cumulus-oocyte complexes were allocated to IVF or ICSI based on sperm analysis after processing the semen. Insemination/injection was never performed more than 40 h post-hCG in order to avoid *in vitro* ageing. According to national reimbursement policy, ICSI percentage was exceptionally high (n = 88; 88%). In all cases of conventional IVF (n = 12; 12%) co-incubation of gametes was shortened to 2 h, after which oocytes were denuded and kept in culture overnight until assessing fertilisation (Le Bras et al., 2017).

Oocytes were transferred immediately after ICSI or short co-incubation with sperm (IVF) to a CulturCoin®, which is the dish specially designed for the Miri® TL time-lapse incubator (Esco Medical). *In vitro* culture in the separate chambers of the Miri® TL incubator was performed under reduced oxygen (5%) with 6.2% CO_2. The volume of sequential culture medium used (OS Cleave™, Origio, Måløv, Denmark) was 30 µL, which was covered with sterile mineral oil (Gynemed, Lensahn, Germany) to protect from evaporation. The recommended t0 (start of video sequence) was either set as the mid-point between the beginning and the end of ICSI or as the time of insemination (IVF).

Images were acquired by a built-in Zeiss objective (x20) specialised for 635 nm illumination using red light. Videos consisted of > 10,000 images (120 h, 7 focal planes, image frequency 5 min). All videos were analysed with the Miri® TL Viewer software (Esco Medical). Annotation was conducted according to previously published guidelines (Ciray et al., 2014) and included t2PB, tPNa, tPnf, t2-t8, tSC, tSB, and tB.

Fertilisation control was performed on the morning of day 1, which is equal to 18-20 h post-injection or insemination. When no signs of fertilisation were observed, Miri TL videos were used to screen for slowly or rapidly developing embryos, e.g. those showing 2 Pn off the above-mentioned time frame.

At cleavage stages (days 2 and 3), all embryos were graded according to blastomere number, size, and uniformity, as well as the presence of fragmentation and multinucleated cells (Alpha Scientists in Reproductive Medicine, ESHRE Special Interest Group of Embryology, 2011). On the morning of day 3, embryos were transferred to fresh, pre-equilibrated 30-µL droplets of OS Blast™ medium (Origio). Day 4 morphology was scored according to in-house guidelines (Ebner et al., 2009a). Static blastocyst morphology, however, was assessed at 116 ± 2 h according to the above-mentioned guidelines (Gardner et al., 2000; Alpha Scientists in Reproductive Medicine, ESHRE Special Interest Group of Embryology, 2011) focusing on expansion grade and the quality of both cell lineages (without removing the dishes from the TL-incubator). Transfer or cryopreservation never occurred earlier than 118 h.

At the blastocyst stage, the whole videos were screened for the presence of cytoplasmic strings bridging the inner cell mass with the trophectoderm (primary objective). For this purpose, all seven focal planes of the Miri TL® were used. As recommended by an expert time-lapse user group (Ciray et al., 2014), the onset of cytoplasmic strings or bridges (triangular- or thread-like) was annotated as tCS (Figure 2, Video 1). In cases where more cytoplasmic strings were observed, tCS referred to the appearance of the first string. VCS is the time period during which the cytoplasmic strings were visible. If there was more than one cytoplasmic string, the disappearance of the last one was used to calculate VCS. In some cases (n = 16), cytoplasmic strings were still observable at the time of embryo transfer. These cycles were not used for calculating the VCS.

Apart from these morphokinetic parameters, blastocyst formation was also analysed for the presence or absence of collapses. Collapses were defined according to criteria proposed by Marcos et al. (2015) based on a total or partial separation of trophectoderm cells from the zona pellucida. Importantly, to be considered a real collapse, a separation of > 50% of the trophectoderm from the zona pellucida had to occur (Marcos et al., 2015). Minor detachments were not analysed in the present study. Number of collapses, but not morphokinetic data, was used for analysis (Bodri et al., 2015).

Since it is not clear whether the frequency and the underlying cause of blastocoelic collapses differs between fresh and vitrified/warmed blastocysts (Shimoda et al., 2016, Ebner et al., 2017a) only those collapses before transfer or cryopreservation were considered.

Vitrification

In all cases, vitrification and warming was performed (Ebner et al., 2009b) using a commercially available kit (VitriStore, Gynemed) based on ethylene glycol and DMSO (for vitrification) as well as sucrose (for warming). Kitazato cryotop (Gynemed) served as a carrier system. No artificial shrinkage of the blastocoel was applied before vitrification, but as a routine, blastocysts (except Gardner score 5) were artificially hatched by means of laser pulses after warming. As an internal guideline, for proper vitrification, the quality of the ICM and TE was not to be worse than quality B in order to optimise cryosurvival. The only exception was if a blastocyst was the only one available or any of the two cell lineages was of quality A (Ebner et al., 2017a). Since cytoplasmic strings were found to be highly susceptible to manipulation (Salas-Vidal et al., 2004), to the strings were not re-identified after warming (i.e., no time-lapse video sequences were acquired for warmed blastocysts).

In total, 95 fresh and 55 vitrified/warmed transfers were performed. Approximately two-thirds of the blastocyst transfers were elective (68%). The presence of strings or collapses was never a decision criterion in terms of selection, but 82 (54.7%) transfers were derived from string-positive blastocysts.

Treatment outcome

As mentioned above, the secondary endpoint of this study was treatment outcome. Biochemical pregnancies were defined as a significant increase in hCG levels two weeks after intrauterine blastocyst transfer (> 10 mU/ml). Four weeks later, embryo transfer implantation was confirmed by ultrasound visualisation. Clinical pregnancies showed at least one gestational sac with positive heart activity. In terms of live-birth, information about birth weight, weeks of gestation, complications during pregnancy (vaginal bleeding, contractions), malformations, and transfer to a neonatal care unit was collected.

Statistics

According to the results of the Kolmogorov-Smirnov test for normal distribution, a chi-square-, a Mann-Whitney-U-, a Kruskal-Wallis-, or a Fisher´s Exact test was used. For univariate analysis of categorical variables, we used chi-square tests. A Mann-Whitney-U-Test and an Independent Samples Median test were performed to compare distributions/medians. The data were analysed using SPSS version 22 (SPSS, Inc., Chicago IL) and a P-value < 0.05 was considered to be statistically significant. All significances were separately calculated for both fresh and vitrified cycles, and since there was no observed difference, data were pooled. This could also be done for IVF and ICSI.

Results

In the fresh cycles of the 100 patients, 1007 cumulus-oocyte complexes were collected (12.0 ± 5.8). After denudation (ICSI) or short co-incubation with sperm (IVF), almost 82% (n = 824) of the complexes did contain a metaphase II oocyte. A total of 620 (75.2%) showed regular signs of fertilisation after IVF or ICSI and were consequently kept for *in vitro* culture up to the blastocyst stage. The associated blastocyst formation rate was 62.4% (387/620). IVF blastocyst formation (67.6%; 50/74) did not differ from ICSI (61.7%; 337/546).

Blastocyst quality

At 116 ± 2 h, one quarter of the blastocysts were at early blastocyst stage, 53 (13.7%) at Gardner grade 1, and 47 (12.1%) at grade 2. A similar proportion developed to full blastocyst stage (88; 22.7%). The majority of the blastocysts was at the expanded stage (173; 44.7%), while another 26 blastocysts (6.7%) already started to hatch out of the zona pellucida (Gardner grade 5).

In those blastocysts of which the inner cell mass and trophectoderm could be scored (Gardner grade 3-5), 91 (31.7%) were of optimum quality, e.g. having both cell lineages scored as A. In another 151 of these blastocysts (52.6%), one or both cell lineages were of quality B. Only 32 of the larger blastocysts (8.3%) did not qualify for transfer or vitrification. The total utilisation rate (transfer or vitrification) of day 5 embryos was 320/387 (82.7%).

Cytoplasmic strings

Interestingly, cytoplasmic strings were never seen at the early blastocyst stage, rather, they formed around the full blastocyst stage (Figure 2). A total of 170 blastocysts, corresponding to 43.9% of all blastocysts and 59.2% of blastocysts grade 3-5, had at least one cytoplasmic string (mean number of strings: 2.0 ± 1.1, range: 1-7). As indicated in Table 2, the presence of strings correlated with grade of expansion ($P < 0.001$). String positivity was comparable between IVF (20/50; 40.0%) and ICSI (150/337; 44.5%).

Approximately two thirds (109/170) of the blastocysts showed a least one string with a bulge (for details see below). The earliest appearance of strings was at 90.8 h, the latest detection was made at 120.4 h (mean time of first observation: 108.5 ± 6.4 h). On an average, cytoplasmic strings were visible over a period of 5.2 ± 3.5 h (VCS), excluding those blastocysts that still showed strings at the time of transfer/cryopreservation. Blastocyst quality was not related to the occurrence of cytoplasmic strings ($P > 0.05$). Fertilisation method was also not related, since there was no difference in this rate between IVF and ICSI. None of the annotated morphokinetic parameters (Figure 3) differed between those blastocysts that showed at least one cytoplasmic string and those that did not ($P > 0.05$).

Collapses

Out of all blastocysts in culture, 151 (39.0%) showed at least one collapse during the process of expansion (mean number of collapses: 1.6 ± 0.9; range: 1-5). There was a significant association between the presence of collapses and cytoplasmic strings ($P < 0.001$). Specifically, 115 blastocysts showed at least one cytoplasmic string and at least one collapse. This figure corresponded to 29.7% of all blastocysts and to 67.7% of all blastocysts ≥ grade 3. Again, the presence of collapses was not related to the mode of fertilisation (insemination or injection).

Treatment outcome

In the 95 fresh cycles, 45 pregnancies (47.4%) were achieved, resulting in a live birth rate of 40% (38/95). Two biochemical pregnancies and 5 missed abortions were recorded. In the subsequent 55 frozen-thawed cycles, the corresponding rates were 41.8% and 32.7% (5 missed abortions). One minor malformation occurred in the fresh cycles (subcutaneous fistula; 1.8% of all births). Female age ($P = 0.016$) was the only patient-related factor that was significantly associated with pregnancy rate. Out of the 3 main parameters of blastocyst morphology, only the grade of expansion was significantly correlated to pregnancy rate ($P = 0.023$) once grades 1 and 2 (early blastocyst stage) were pooled. This significance was lost for live birth.

The presence of cytoplasmic strings did not affect pregnancy rate ($P > 0.05$), neither did VCS ($P > 0.05$). The detection of cytoplasmic strings showed an insignificant trend of being a positive predictor of live birth ($P = 0.086$), with 92% (35/38) of all string-positive pregnancies (Table 3) resulting in live birth (70% (21/30) in the string-negative counterparts).

Table 3 also highlights neonatal outcome for different blastocyst qualities, and it should be mentioned that multiple entries were possible. Neither blastocyst quality according to Gardner criteria nor supplementary analysis on the basis of strings and collapses was associated with neonatal outcome (sex, birth weight, height, week of gestation, malformations).

Discussion

Apart from known benefits, such as uninterrupted embryo culture or 24 h-surveillance, time-lapse technology offers the potential to refine our knowledge on preimplantation morphology and its theoretical capability as a predictive tool. A group of specialists (Ciray et al., 2014) was the first to try to standardise nomenclature and annotation process, in particular. These authors set standards for defining special timings and how to calculate dynamic embryo cleavage variables. They further suggested additional annotations (e.g., fragments, vacuoles, clusters of smooth endoplasmic reticulum) and they strongly recommended screening for irregular cleavage events.

Amongst the latter group, the formation and presence of cytoplasmic strings at the blastocyst stage is of particular interest, since information on this dysmorphism is scarce. Scott (2000) was the first to note those cytoplasmic “processes that extend as tongue-like structures from the trophectoderm cells to a central area on the inner cell mass” in human blastocysts. Such cytoplasmic strings begin to arise at the end of early blastocyst stage (Ciray et al., 2014) in the form of rather triangle-shaped structures, and as the expansion proceeds (Gardner, 2000), they change their appearance towards more distinct string-like projections. It indeed seems that width and length of the cytoplasmic structures vary with expansion grade (Salas-Vidal and Lomelí, 2004). The present study did not focus on triangle-shaped precursor extensions, which explains the later onset of strings in the present dataset. It has also been claimed that cytoplasmic strings are transitional and that their persistence in expanded or hatching blastocysts was previously deemed a negative predictor, being the result of polarisation problems or poor media conditions (Scott, 2000; Hardarson et al., 2012).

The idea that persistent cytoplasmic strings are an *in vitro* artefact is supported by murine data. Salas-Vidal and Lomelí (2004) reported that cytoplasmic filopodia were literally seen in all *in vitro* cultured blastocysts, but only in 40% of the *in vivo* developed ones. Whether this divergence between *in vitro*- and *in vivo*-generated blastocysts also holds true in humans is yet to be determined. No question that cytoplasmic strings are omnipresent in human ICSI/IVF-blastocysts (and the present time-lapse study shows that the incidence of this morphological parameter was relatively underestimated in the past) but for obvious reasons, information on *in vivo* counterparts is scarce. It is noteworthy that at least some *in vivo* conceived human blastocysts, which were flushed from uteri of volunteers, did show cytoplasmic strings (Munné et al., 2018).

The close association between the presence of strings and blastocoelic collapses found in the present study provides the first evidence that the formation of at least some type of strings is artificial and potentially associated with one or more collapses. This morphokinetic phenomenon is well described for human *in vitro* blastocysts and is defined as a separation of the majority of TE cells from the zona pellucida during blastocyst growth (Marcos et al., 2015; Bodri et al., 2016) and blastocyst hatching (Kovacic et al., 2018).

Previous studies showed that the implantation potential is limited in the presence of collapses and they recommended not to transfer affected blastocysts if alternatives are available (Marcos et al., 2015). Recently, Bodri et al. (2016) calculated that collapses are not independent predictors of live birth since they are confounded by stronger predictors such as female age. Although this was not the main focus of our study, our data confirm the findings of Bodri et al. (2016). We did find that in contrast to female age, collapses are not a predictor of pregnancy or live birth. Hypothetically, a series of collapses that artificially generates cytoplasmic strings could eventually lead to monozygotic twinning if the inner cell mass is mechanically stressed by the contractions, especially if the ICM is of poor quality or loose cohesiveness (Otsuki et al., 2016).

Although there is consensus that strings at later stages of blastocyst development, particularly persistent ones, may reflect some underlying defect (Scott, 2000; Hardarson et al., 2012), other researchers think that this supposed anomaly does not affect blastocyst viability (Ciray et al., 2014; Alpha Scientists in Reproductive Medicine, ESHRE Special Interest Group of Embryology, 2011). Notably, the mere existence of cytoplasmic strings after blastulation is considered to be physiological. The occurrence of such projections in the phase of blastocyst expansion is a prerequisite for a controlled cell egress from the polar to the mural trophectoderm (Gardner, 1996).

An earlier hypothesis (Ducibella et al., 1975) suggested that the observed cytoplasmic extensions could fix the ICM to one pole of the embryo, thus isolating it from the blastocoel. Alternatively, those cytoplasmic strings were thought to enduringly anchor the initial trophectoderm cell in the polar-mural junction (Gardner, 1996). Although partly obsolete today, the said theories stressed that the ICM and trophoblast communicate with each other (Dulcibella et al., 1975). In fact, these strings are predominantly present in junctional trophectoderm cells which make up the boundary between the polar (overlying the inner cell mass) and the mural region of trophectoderm. The cytoplasmic strings are directed towards the surface of the ICM and, as highlighted above, are involved in the polarised flow of cells from the polar to the mural trophectoderm. As the cells reach their final location, the cytoplasmic strings tend to withdraw. This withdrawal theoretically could lead to the interpretation that some cytoplasmic strings fail to reach the ICM surface (Gardner, 2000).

Using a mouse model, Salas-Vidal and Lomeli (2004) shed further light on the actual interaction between both cell lineages. These authors identified three different types of cellular projections, long traversing cytoplasmic filaments connecting ICM and mural trophectoderm (major portion of the present data set), short filopodia originating from either of the cell lineages that protruded into the blastocoel (considerably underrepresented here) and those related to junctional TE-cells (Fleming et al., 1984). It is of special interest that no short filopodia protruding into the blastocoel could be detected in the present study and it remains unclear whether these structures are not present in human blastocysts or could not be identified due to the lower magnification (x20) currently used in all time-lapse systems available. This diversity and the pooling of cytoplasmic strings of different aetiology could be one explanation for the current lack of knowledge in terms of implantation behaviour (Scott, 2000; Alpha Scientists in Reproductive Medicine, ESHRE Special Interest Group of Embryology, 2011; Hardarson et al., 2012, Ciray et al., 2014).

As one would expect for the cytoplasmic strings that are involved in cell migration, the cytoskeleton of these structures was found to be actin-rich (Salas Vidal and Lomelí, 2004). Cytochalasin B, a known inhibitor of actin polymerisation, eliminated all types of cytoplasmic extensions except short (actin-free) filopodia (Salas Vidal and Lomelí, 2004). More interestingly, the same group (Salas Vidal and Lomelí, 2004) provided immunofluorescence evidence that strongly suggested a direct communication between ICM and mural trophectoderm cells. In detail, they referred to two receptors, fibroblast growth factor receptor 2 and human epidermal growth factor receptor 3, both of which play an important role in embryonic development and signal transduction, respectively. The specific distribution of these molecules suggests that they are not passively localised but rather found in specialised cell compartments of the cell projections. Indeed, in mice, bulges were observed in such cytoplasmic structures (Salas Vidal and Lomelí, 2004) moving towards the cell body which could be seen as the manifestation of a vesicle transport mechanism (Miller et al., 1995). Although detecting such vesicles was not the main subject of the present analysis, it should be mentioned that these structures also exist in human blastocysts. Australian authors did find a higher number of such “travelling” vesicles (figure 2) in human blastocysts that led to clinical pregnancy (Eastwick et al., 2019).

Conclusion

To summarise, this study provides the first evidence that the mere presence of cytoplasmic extensions on day 5 is not a negative predictor of outcome, but rather is positively correlated with clinical pregnancy (Eastwick et al., 2019). The significantly higher expression of cytoplasmic strings in blastocysts of optimal quality would further support this finding (Table 2). The interpretation of string-related data may be complicated by the fact that obviously two entities of cytoplasmic extensions exist in IVF: physiological (e.g. showing bulges) and artificial ones (being associated with blastocoelic collapses). To additionally complicate matters, the simultaneous presence of both types of strings is a fact (Figure 2). To support the hypothesis of co-existence of different types of cytoplasmic strings, further analysis of the cytoskeleton of presumed artificial strings is needed (no signs of signal transduction would be expected), which is particularly difficult in human blastocysts.

Video 1. Time-lapse sequence of a blastocyst (321) that showed a collapse at 117.0 h followed by the formation of small cytoplasmic strings (117.5 h; 12 o´clock position). Scoring was performed according to Alpha Scientists in Reproductive Medicine and ESHRE Special Interest Group of Embryology (2011).

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Key message

Time-lapse analysis of cytoplasmic processes at blastocyst stage reveals that this morphological feature is physiological and does not interfere with clinical pregnancy and live birth. Some strings are an artefact potentially being accociated with blastocoelic collapses.

Figure 1. Schematic presentation of the study design.

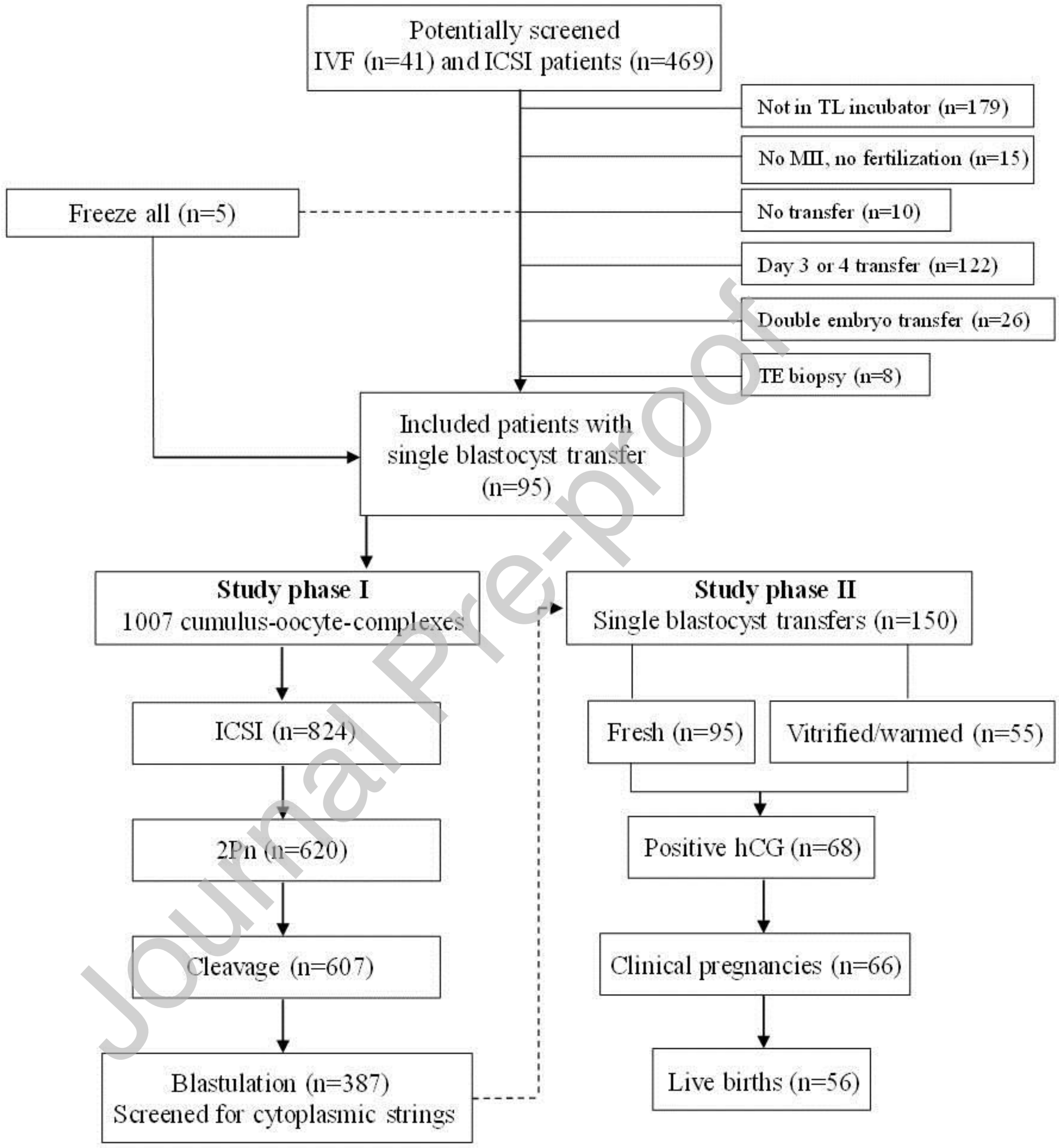

Figure 2. String-positive blastocyst that led to the birth of a healthy girl. BC: blastocoel; CS: cytoplasmic string; ICM: inner cell mass; jTE: junctional trophectoderm; mTE: mural trophectoderm; tCS: triangular-shaped cytoplasmic string; tV: travelling vesicle.

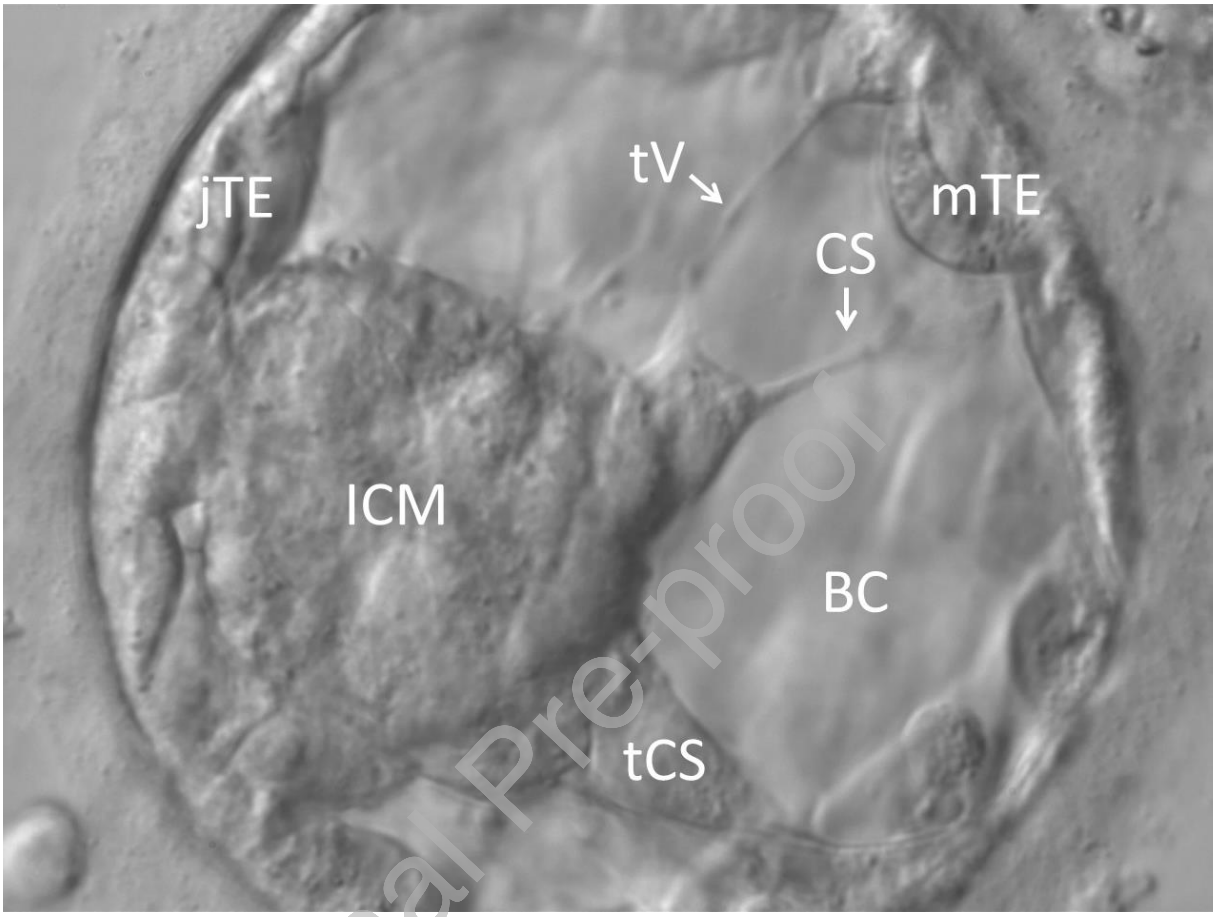

Figure 3. Morphokinetic comparison between blastocysts with and without strings.

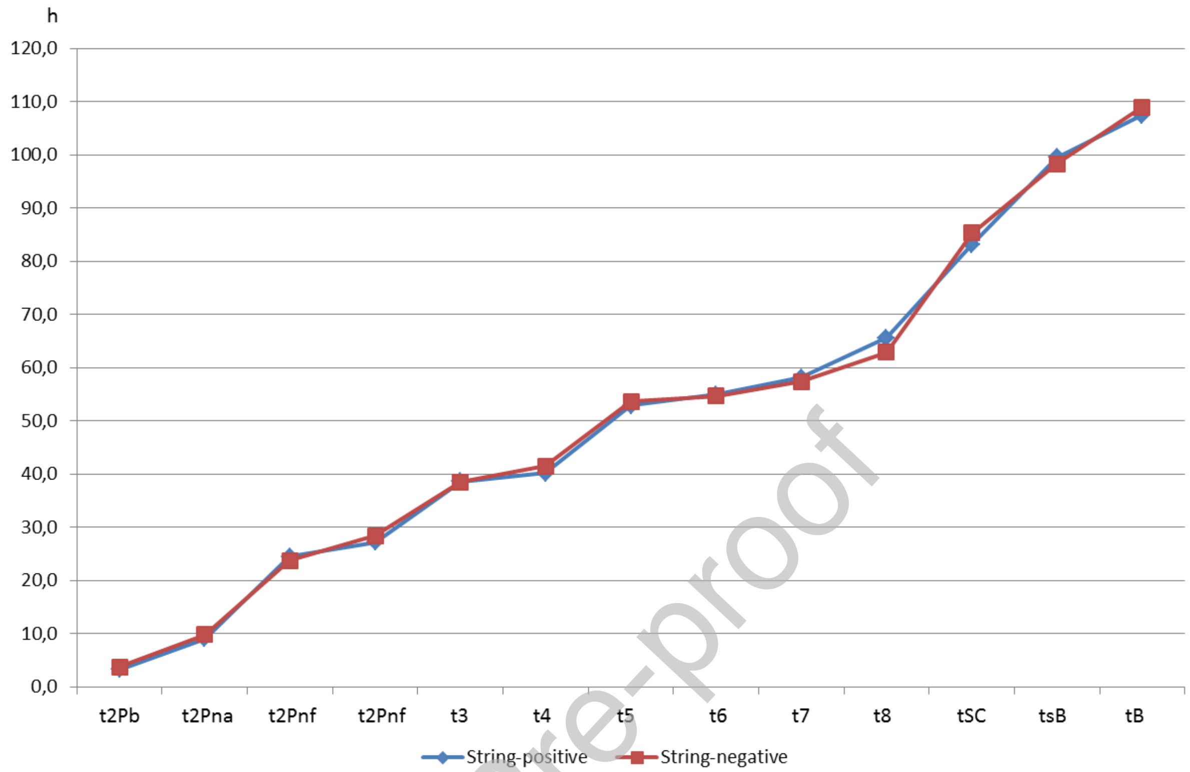

Table 1. Patient and stimulation data in both transfers deriving from string positive and string-negative blastocysts.

	String-positive transfers	**String-negative trans**
N	82	68
Age	32.9±4.8	32.5±4.4
AMH (ng/mL)	3.4±1.3	3.6±1.6
Basal FSH (IU/L)	6.3±1.3	6.4±1.8
Basal LH (IU/L)	5.6±3.1	4.8±2.7
Primary infertility	63%	62%
Antagonist protocol	80%	85%
Dosage of gonadotropins (IU/L)	1850±1155	1720±1095
Duration of stimulation (d)	10.5±1.8	10.4±1.9
Estradiol at time of OI (pg/mL)	1823±1040	1938±1177
Progesterone at time of OI (ng/ml	0.61±0.26	0.64±0.37
LH at time of OI (IU/L)	1.6±1.4	1.8±1.6
Endometrium (mm)	10.4±1.9	10.0±2.0

OI: ovulation induction

Table 2. The abundance of cytoplasmic strings and blastocoel collapses in blastocysts of various expansion stages and quality (n=387).

Cytoplasmic strings	Blastocoelic collapses			
n (%)	tCS	VCS	n (%)	
n	170	151		
Expansion				
Early	0			2/100 (2.(
Full	36/88 (40.9)[a]	110.8.9±6.4	3.3±1.8	43/88 (48.9)
≥Expanded	134/199 (67.3)	107.9±6.2	5.7±3.7	106/199 (53.3)
Quality				
AA	68/91 (74.7)[b,c]	107.5±6.0	6.7±4.0	49/91 (53.8)
AB, BA, BB	76/151 (50.3)[b]	109.1±6.8	4.1±2.6	77/151 (51.0)
<BB	26/45 (57.8)[c]	109.4±6.0	4.8±2.6	26/45 (57.8)

[a,b]$P<0.001$

[c]$P<0.05$

tCS = onset of cytoplasmic strings; VCS = visibility of cytoplasmic strings.

Table 3. Presence of strings and its association with blastocyst quality and treatment/neonatal outcome in single blastocyst transfers (n=150).

	String-positive blastocysts	String-negative blastocysts
n	82	68
Expansion		
Early	0	18 (26.5)
Full	16 (19.5)	18 (26.5)
≥Expanded	66 (80.5)	32 (47.0)
Quality		
AA	43 (52.4)	15/50 (30.0)
AB, BA, BB	34 (41.5)	31/50 (62.0)
<BB	5 (6.1)	4/50 (8.0)
Collapse[a]	50 (61.0)	13 (19.1)
Positive hCG	38 (46.3)	30 (44.1)
Clinical pregnancy	38 (46.3)	28 (41.2)
Live birth	35 (46.3)	21 (26.5)
Week of gestation	39.6±1.6	39.8±1.1
Sex ratio	0.6	0.8
Birth weight	3398±453	3304±344
Height	51.4±2.4	51.2±2.1
Malformations	1 (2.9)	0

[a] P<0.001.